Different methods for gradient descent stepsize choice

Let us assume that we are given a function $f\colon \mathbb{R}^d \to \mathbb{R}$ which is L-smooth, μ-strongly convex, and twice continuously differentiable, i.e., $f \in C^2(\mathbb{R}^d)$. As a reminder, Gradient Descent (GD) is an iterative method for finding the minimum of a function, and it can be described as follows.

Algorithm 1 GD

1: **Parameters:** stepsize $\gamma > 0$, starting point $x_0 \in \mathbb{R}^d$
2: **for** $k = 0, 1, 2, \dots$ **do**
3: Set $g_k = \nabla f(x_k)$
4: $x_{k+1} = x_k - \gamma \nabla f(x_k)$

Our aim is to choose the stepsize γ so that GD converges "faster". The connection between the stepsize and the convergence speed can be described by the following lemma.

Lemma 1. *Assume f is L-smooth and μ-strongly convex. Then the iterates of GD satisfy the following inequality for all $\gamma \geq 0$ and all $k \geq 0$:*

$$\|x_{k+1} - x^*\|^2 \leq (1 - \gamma\mu)\,\|x_k - x^*\|^2 - 2\gamma(1 - \gamma L)\, D_f(x_k, x^*), \tag{1}$$

where $x^* \overset{def}{=} \arg\min_{x \in \mathbb{R}^d} f(x)$ and $D_f(x, y)$ is the Bregman divergence, defined as follows

$$D_f(x, y) \overset{def}{=} f(x) - f(y) - \langle \nabla f(y), x - y \rangle \qquad \forall x, y \in \mathbb{R}^d.$$

Note that $D_f(x, y) \geq 0$ for every convex function f, and hence for every μ-strongly convex function.

Proof of Lemma 1.

$$\begin{aligned}
\|x_{k+1} - x^*\|^2 &= \big\|\big(x_k - \gamma \nabla f(x_k)\big) - \big(x^* - \gamma \nabla f(x^*)\big)\big\|^2 \\
&= \big\|(x_k - x^*) - \gamma\big(\nabla f(x_k) - \nabla f(x^*)\big)\big\|^2 \\
&= \|x_k - x^*\|^2 + \gamma^2 \|\nabla f(x_k) - \nabla f(x^*)\|^2 - 2\gamma \langle x_k - x^*, \nabla f(x_k) - \nabla f(x^*) \rangle .
\end{aligned}$$

From strong convexity we obtain

$$\langle x_k - x^*, \nabla f(x_k) - \nabla f(x^*) \rangle \geq D_f(x_k, x^*) + \frac{\mu}{2} \|x_k - x^*\|^2 .$$

Also, from L-smoothness we have

$$\|\nabla f(x_k) - \nabla f(x^*)\|^2 \leq 2L\, D_f(x_k, x^*),$$

and, as a result,

$$\|x_{k+1} - x^*\|^2 \leq (1 - \gamma\mu)\,\|x_k - x^*\|^2 - 2\gamma(1 - \gamma L)\, D_f(x_k, x^*),$$

which finalizes the proof. □

As mentioned above, $D_f(x_k, x^*) \geq 0$. Hence, if we take $0 \leq \gamma \leq 1/L$, then $1 - \gamma L \geq 0$ and the last term in (1) is non-positive. Dropping it and applying (1) k times, we obtain the following theorem.

Theorem 2 (Convergence of GD)**.** *Assume f is L-smooth and μ-strongly convex. Then the iterates of GD with stepsize $0 \leq \gamma \leq 1/L$ satisfy*

$$\|x_k - x^*\|^2 \leq (1 - \gamma\mu)^k \|x_0 - x^*\|^2 . \tag{2}$$

To make this result more concrete, we may ask how many GD iterations it takes for the error to reach a certain target level ε. To answer this, one can start from inequality (2) and show that with $\gamma = 1/L$, for every k that satisfies

$$k \geq \frac{L}{\mu} \log \frac{1}{\varepsilon}$$

we have

$$\|x_k - x^*\|^2 \leq \varepsilon \|x_0 - x^*\|^2 .$$

What we can conclude is that if the function is L-smooth and μ-strongly convex, then we can choose some $0 \leq \gamma \leq 1/L$ in **advance** for all iterations of GD, and we can guarantee that after a certain number of steps the squared distance between the current point and the minimizer is reduced by a factor of $\frac{1}{\varepsilon}$ compared to the initial point.

But what if we are allowed to change the stepsize during the algorithm, that is, to pick a new stepsize γ_k at each step k? If we are willing to do this "extra" work, and of course if we are able to do it, can we find stepsizes γ_k that are better than a constant one? Or at least not worse, in the sense of the ε-guarantee from the previous part?

It turns out that the answer is yes, and one such method is called the Polyak stepsize.

Consider a variant of GD with Polyak stepsize, i.e., with the adaptive (i.e., iteration-dependent) stepsize rule

$$\gamma_k = \frac{2D_f(x_k, x^*)}{\|\nabla f(x_k) - \nabla f(x^*)\|^2},$$

where we assume that $\nabla f(x_k) \neq \nabla f(x^*)$. This is a very natural assumption: since $\nabla f(x^*) = 0$ and the minimizer of a strongly convex function is unique, so $\nabla f(x_k) = \nabla f(x^*)$ would mean that $x_k = x^*$, i.e., we have already found the minimum.

(i) From L-smoothness it can be obtained that $\gamma_k \geq 1/L$. That is, the Polyak stepsize is always at least as large as the largest constant stepsize $\gamma = 1/L$ allowed by Theorem 2.

(ii) From μ-strong convexity it can be obtained that $\gamma_k \leq 1/\mu$.

(iii) Finally, it can be shown that after one step of GD with Polyak stepsize we have

$$\|x_{k+1} - x^*\|^2 \leq (1 - \gamma_k\mu) \|x_k - x^*\|^2 .$$

Indeed, let us start from the GD step $x_{k+1} = x_k - \gamma_k \nabla f(x_k)$. Since $\nabla f(x^*) = 0$, we can subtract the zero term $\gamma_k \nabla f(x^*)$ without changing anything, and then group the terms:

$$\|x_{k+1} - x^*\|^2 = \big\|\big(x_k - \gamma_k \nabla f(x_k)\big) - \big(x^* - \gamma_k \nabla f(x^*)\big)\big\|^2 = \big\|(x_k - x^*) - \gamma_k\big(\nabla f(x_k) - \nabla f(x^*)\big)\big\|^2 .$$

Expanding the square and taking γ_k out of the last two terms, we obtain

$$\|x_{k+1} - x^*\|^2 = \|x_k - x^*\|^2 + \gamma_k \Big(\gamma_k \|\nabla f(x_k) - \nabla f(x^*)\|^2 - 2 \langle x_k - x^*, \nabla f(x_k) - \nabla f(x^*) \rangle \Big).$$

Now we bound the inner product from below. By μ-strong convexity,

$$\langle x_k - x^*, \nabla f(x_k) - \nabla f(x^*) \rangle \geq D_f(x_k, x^*) + \frac{\mu}{2} \|x_k - x^*\|^2 .$$

The inner product enters with a minus sign and $\gamma_k \geq 0$, so replacing it by this lower bound can only increase the right-hand side. Moreover, in front of the squared gradient norm we substitute the definition of γ_k. This gives

$$\|x_{k+1} - x^*\|^2 \leq \|x_k - x^*\|^2 + \gamma_k \left(\frac{2D_f(x_k, x^*)}{\|\nabla f(x_k) - \nabla f(x^*)\|^2} \cdot \|\nabla f(x_k) - \nabla f(x^*)\|^2 - 2D_f(x_k, x^*) - \mu \|x_k - x^*\|^2 \right).$$

Here the squared gradient norms cancel, so the first term in the brackets is just $2D_f(x_k, x^*)$, and it cancels with $-2D_f(x_k, x^*)$. Only the term $-\mu \|x_k - x^*\|^2$ remains, and therefore

$$\|x_{k+1} - x^*\|^2 \leq \|x_k - x^*\|^2 - \gamma_k \mu \|x_k - x^*\|^2 = (1 - \gamma_k \mu) \|x_k - x^*\|^2 ,$$

which is exactly the inequality in (iii).